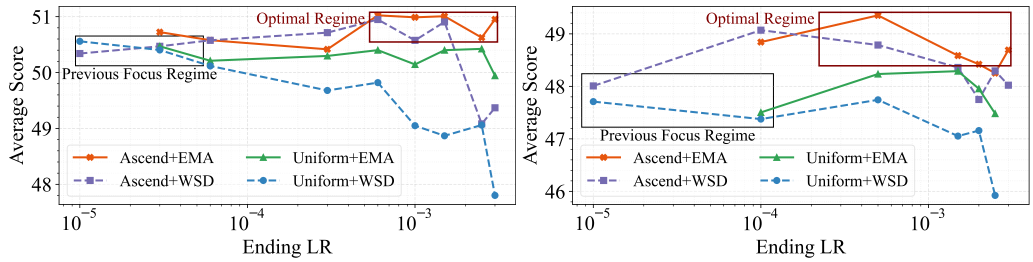

(a) DCLM. (b) Mid-Training (WebOrganizer-DCLM).

Figure 5: This figure compares various training strategies, identifying a high-performing and previously underexplored **Optimal Regime** where moderate learning rate (LR) decay, weight averaging, and curriculum learning produce synergistic advantages. We run experiments on both Uniform (uniformly ordered data) and Ascend (training data arranged by ascending DCLM scores) data schedules. For both schedules, we conduct an ablation on the ending learning rates of WSD schedules, ranging from 1×10^{-5} to 1×10^{-3}, representing aggressive to moderate LR decay. We denote strategies applying weight averaging as *EMA*, which compute the final model checkpoint via an EMA of the last six checkpoints, and denote those strategies without weight averaging as *WSD*. We measure performance by the average downstream task score (as in Table 1). This newly identified regime contrasts with the **Previous Focus Regime**, which represents common practices without a data curriculum or weight averaging, and with an ending LR between 1×10^{-5} and 1×10^{-4}. This range is typical in prior work, which often uses an ending LR of one-tenth of a peak LR (on the scale of $\times 10^{-4}$) (Dubey et al., 2024; DeepSeek-AI et al., 2024) or fixes the ending LR around 10^{-5} (Li et al., 2024; 2025b). This observation also holds for mid-training settings.

cay (i.e., a higher ending LR than in standard practice). While a data curriculum with moderate LR decay alone also achieves near-optimal results, it does not fully match the combined strategy and may need a more careful tuning of ending LRs. These two curriculum-based strategies both outperform their uniform-ordering training counterparts when the LR decay is properly tuned for both curriculum-based and uniform-ordering. Furthermore, as shown in Figure 5(b) of the mid-training setting, the best combination can improve by 1.68% in the average benchmark accuracy over the baseline (Uniform+WSD with ending LR 1×10^{-5}, corresponding to the left endpoint of the blue dash line), a prominent improvement without any additional data refinement. This motivates the following guideline for curriculum-based pretraining: *use a moderate LR decay and adopt model averaging*. To distinguish this from CMA, we call this strategy **C**urriculum with LR **D**ecay **M**odel **A**veraging (**CDMA**). Specifically, curriculum-based LLM pretraining should use a more moderate LR decay than the optimal setting for uniform data training; their optimal regimes can differ substantially (around 1×10^{-5} for uniform training versus roughly 1×10^{-3} for curriculum-based training in our setting). A weight averaging strategy can further enhance performance and produce more stable benefits in this moderate LR regime. A more systematic recipe for the strategy combinations can be a promising direction for future work.

Discussion: Why is this Combination Under-Explored? The CDMA strategy is straightforward and effective, which raises the question of why it has been underexplored. A possible explanation is that prior work has focused on an aggressive LR decay regime, which has obscured the discovery of alternative approaches. Confirmed by prior work (Li et al., 2025b), this aggressive decay regime is close to optimal for the standard uniform data scenario, and there is a clear trend favoring a near-zero ending LR, as shown in Figure 5. However, the optimal regime for uniform data is not necessarily optimal for other settings. Prior work focusing on curriculum design mostly adopts cosine annealing schedules, within this aggressive decay regime, and has consequently reported marginal or disappointing results (Zhang et al., 2025). The negative results caused by these compounding factors can prevent more progress on curriculum-based pretraining. Hence, our proposed optimization space involving the co-design of LR schedules, data curricula, and model averaging strategies is still underexplored

Ablation Study. We conduct ablation studies to verify the robustness of our method. Our approach generalizes effectively when evaluated with an alternative quality metric (PreSelect) and a different, unfiltered pretraining dataset (WebOrganizer) (SHUM et al., 2025; Wettig et al., 2025). Full experimental details are presented in Section A.3.

6 A Theoretical Demonstration

As we reported and discussed above, the benefit of curriculum learning emerges when we apply a weight averaging method instead of a learning rate schedule with excessive decay, such as Cosine

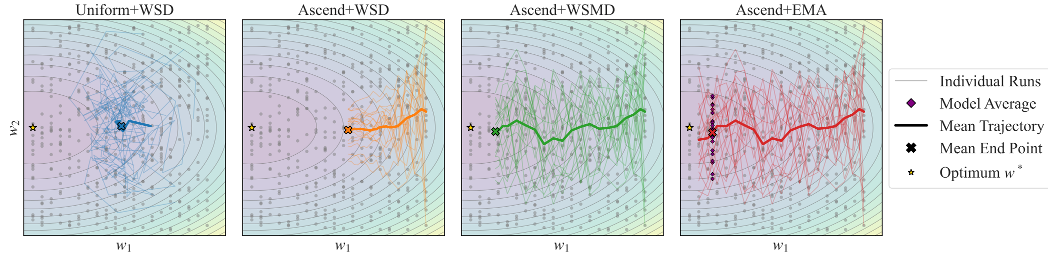

Figure 6: Visualization of the simulation experiments of the theoretical example. The mean trajectory is averaged over $R = 20$ runs. The yellow star marks the global optimal, and w_1 represents a signal direction and w_2 represents a noise direction. The data samples are distributed evenly along the signal direction and randomly located along the noise direction. *Ascend+WSMD* and *Ascend+EMA* win by sufficient progress along signal direction; *Uniform+WSD* fails for inconsistent signal and thus large variance along signal direction; *Ascend+WSD* fails for early-decay, resulting in insufficient update along the signal direction.

or WSD schedules, in practical pretraining. In the following, we present a simple theoretical model that recovers the above empirical insight. The main proof of this section can be found in Section C.

Problem Setup. We consider a quadratic loss function $\mathcal{L}(\boldsymbol{w}) = \frac{1}{2}\|\boldsymbol{w} - \boldsymbol{w}^*\|_2^2$, where $\boldsymbol{w} = (w_1, w_2) \in \mathbb{R}^2$ represents the trainable parameter, and $\boldsymbol{w}^*$ denotes the ground truth, which is set to the original point $(0, 0)$. We use Stochastic Gradient Descent (SGD) to optimize this problem. We denote the one-sample loss used to calculate the gradient for the t-th iteration as $\ell_t(\boldsymbol{w}) := \|\boldsymbol{w} - \boldsymbol{x}_t\|_2^2$, where data point $\boldsymbol{x}_t$ is sampled from some given dataset $\mathcal{D} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(M)}\}$. Therefore, we have the SGD update rule for the t-th iteration as $\boldsymbol{w}_t = \boldsymbol{w}_{t-1} - \eta_t \nabla \ell_t(\boldsymbol{w}_{t-1})$, where the η_t denotes the learning rate in the t-th iteration. We initialize the parameter as $\boldsymbol{w}_0 = (L, 0)$. We denote the learning rate schedule by $E := \{\eta_1, \eta_2, \ldots, \eta_M\}$. We denote $\boldsymbol{w}_t = (w_t^{(1)}, w_t^{(2)})$. We define $\mathcal{W}_{M;E}$ to be the distribution of $\boldsymbol{w}_M$. The randomness within $\boldsymbol{w}_M$ comes from the random draw of the distribution in SGD. We further define the expected loss $\bar{\mathcal{L}}(M; E) := \mathbb{E}_{\boldsymbol{w} \sim \mathcal{W}_{M;E}}[\mathcal{L}(\boldsymbol{w})]$.

In the following, we consider the training dataset $\mathcal{D}$, which consists of M different data points with varying data qualities. Specifically, data point $\boldsymbol{x}^{(i)} = (x_1^{(i)}, x_2^{(i)})$ satisfy that $x_1^{(i)} = (i-1)d$ and $x_2^{(i)} \sim \text{Uniform}(-L, L)$, we further set $d = L/M$. $\boldsymbol{x}^{(i)}$ provides a signal in the first dimension and introduces noise in the second dimension. Next, we consider two sampling strategies for each iteration of SGD: (1) We sample one data point uniformly from $\text{Uniform}(\mathcal{D})$; (2) we sample one data point from $\mathcal{D}$ in an ascending order. In other word, in t-th iteration, $\boldsymbol{x}_t = \boldsymbol{x}^{(M-t+1)} \in \mathcal{D}$. The visualization of optimization trajectories in a simulation experiment can refer to Figure 6.

Uniform Sampling + Learning Rate Schedule. SGD acts as an exponential averaging of the current parameter and the sampled data point. Once we uniformly sample data points with no ordering, then on the x-axis, the parameter would approximately oscillate from 0 to $(m-1)d$ with a large variance. We can prove that given any data schedule E starting with some $\eta_1 \le 1$, the expected loss for uniformly sampling SGD has a lower bound

$$\min_E \bar{\mathcal{L}}(M; E) = \Omega(L^2). \tag{1}$$

This lower bound is derived from the loss on the x-axis. When we apply the uniform sampling, SGD cannot get enough signal towards the right direction; instead, the SGD optimizer would approach the mean of $x_1^{(1)}, x_1^{(2)}, \ldots, x_1^{(M)}$ in expectation.

Ascending Data-Ordering + Practical WSD Schedule. Next, we sample data in an ascending order from $\boldsymbol{x}^{(M)}$ to $\boldsymbol{x}^{(1)}$, using the following WSD learning rate schedule $\tilde{E}$ such that $\eta_t = \frac{1}{2}$ for $1 \le t \le \lfloor 0.9M \rfloor + 1$, and $\eta_t = \frac{1}{T-(M-T_0)}$ for $\lfloor 0.9M \rfloor + 2 \le t \le M$, where $T_0 = M - \lfloor 0.9M \rfloor$. In this learning rate schedule, we follow a practical setting, where we decay 10% of the total iterations. We then show that for this learning rate schedule $\tilde{E}$, the expected loss still cannot break the lower bound

$$\bar{\mathcal{L}}(M; \tilde{E}) = \Theta(L^2). \tag{2}$$

Ascending Data-Ordering + WSMD Schedule. In the above, we show that even using an ascending data-ordering, the loss lower bound does not improve if we decay too much in the learning rate schedule. Next, we show a Warmup-Stable-Moderate Decay (WSMD) schedule with less decay and a larger ending learning rate can better utilize the ascending data-ordering and get a smaller loss. Specifically, do a modification of the above WSD schedule, setting $T_0 = \Theta(M^{\frac{2}{3}})$. WSMD schedule can break through the above lower bound, denoted as E^*

$$\bar{\mathcal{L}}(M; E^*) = \Theta(M^{-\frac{2}{3}} L^2). \tag{3}$$

Ascending Data-Ordering + Stochastic Weight Averaging (SWA). Despite the failure of the practical WSD learning rate schedule, we demonstrate that with a constant learning rate, a sample SWA surpasses the aforementioned lower bound. The reason is that: (1) First, along the x-axis, with a constant learning rate, the updated parameter gets a larger gradient accumulation towards the ground truth compared with the practical WSD with 10% decay, which is too much to get enough loss reduction. (2) Second, the SWA allows appropriate averaging along the y-axis and results in noise reduction, thus leading to smaller loss, as the WSMD schedule does.

Theorem 6.1. *Given a learning rate $\eta_0 \leq 1$, the parameter derived by the averaging on the last n weights $\bar{\boldsymbol{w}}_M = \frac{1}{n}\sum_{t=0}^{n-1} \boldsymbol{w}_{M-t}$, where $n = \Theta(M^{\frac{2}{3}})$ such that the expected loss*

$$\mathbb{E}[\mathcal{L}(\bar{\boldsymbol{w}}_M)] = \tilde{O}(M^{-\frac{2}{3}} L^2),$$

where $\tilde{O}(\cdot)$ hides log factors and constants independent of L and M.

7 Conclusion

In this paper, we investigate the interaction between data schedules and learning-rate (LR) schedules in language model pretraining and identify a fundamental tension between curriculum learning and conventional LR-decay strategies. To address this mismatch, we propose replacing sharp LR decay with a combination of moderate decay and model averaging. This design uncovers a previously underexplored optimization regime that better aligns curriculum-based data ordering with LR decay. Our findings highlight the importance of co-designing LR and data schedules rather than optimizing them in isolation. We advocate for future work on developing more principled and efficient methods for jointly tuning these hyperparameters and coordinating the interplay between training dynamics and distributional shifts—an effort that we believe will be central to further improving the efficiency and effectiveness of LLM pretraining.

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